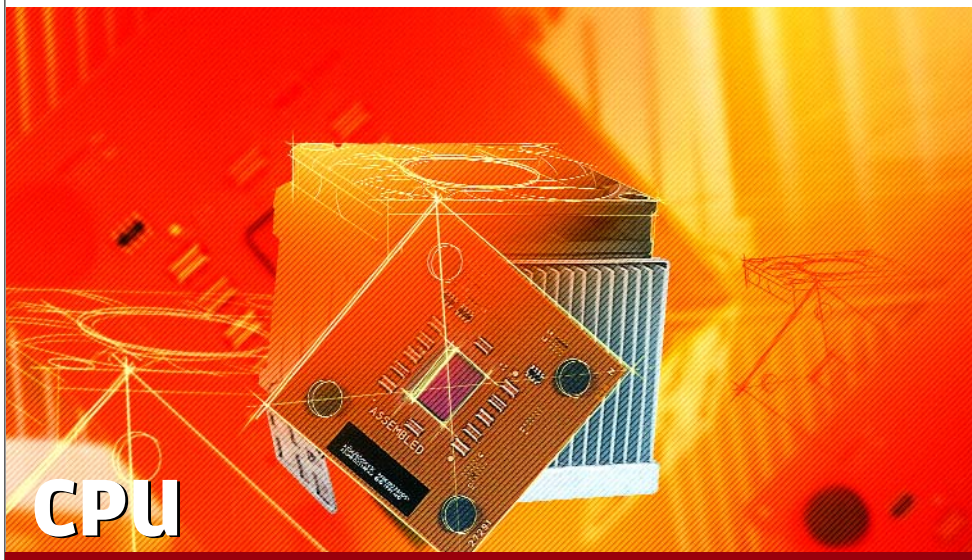




Intel is expected to manufacture its Xeon, and AMD its Opteron, processors using the 90nm technology in late 2005. This is good news, since desktops these days need to work with the computing power of workstations.

However, there are areas of concern as newer processors with newer architectures are released. The upgrade path, for example, has become more difficult than it used to be. Newer CPUs will need mainboards that have chipsets specifically designed for them. A good example is the Intel 3.46 GHz CPU, which supports 800 MHz FSB. This will not work at 800 MHz FSB on older boards and clock down to 533 MHz. Hence, you may have to do a complete overhaul of your system.

AMD plans to manufacture its Opteron processors featuring x86-64 technology. It's all set to have dual-core processing as its prime feature, akin to Intel's Xeon "Tulsa" chip.

Of the two, AMD may have a manufacturing-cost benefit as its CPU cores are smaller.

This, however, does not necessarily mean that AMD offers better performance.

The Facts

☐ A CPU contains many ground pins in order to reduce noise and to prevent it from picking up stray voltages.

☐ According to Andy Moore, the founder of Intel, the number of capacitors on a chip would double every 18 months. Currently, this is not holding true—the latest Pentium 4 CPUs have just marginally more than those released last year.

☐ The 64-bit architecture, which was originally deployed for server applications, is now powering mid-range to high-end desktop systems.

☐ As the average temperature of the CPU rises by 10 degrees, its life-span is halved. This happens over a period of time due to a phenomenon called electromigration. Glitches and system crashes

Ready Reckoner	
 You Need	Decent performance in office-productivity applications
 You Need	Any processor with a speed of 2.0 GHz
 You Need	Any CPU within the range of 2.4 GHz to 2.8GHz
 You Need	Media-editing, home entertainment, and gaming
 You Need	A processor with a speed of 3.0 GHz to 3.4 GHz



are the immediate short term effects of overheating.

Do Remember

1. If you use your PC just for simple office work, it makes no sense in opting for the latest processor. Settle for an AMD sempron or Intel Celeron processor.
2. Graphic designers should look-out for processor with hefty cache memory, as it will make all the difference in your day to day work. Such people should go in for AMD's 64bit processors as the memory controller is also on die and hence reduces the memory sub-system latencies, thus improving overall speeds.
3. Pay special attention towards cooling the processor, as heating might reduce the life span of the processor.
4. The space near the processor in your cabinet should be devoid of cables for free air flow.
5. As of today all processors are available in box packaging with hologram seals, don't not settle for processor with box, as it might be used and devoid of any warranty.



Buying Tips

- ✓ A few 64-bit Linux OS offerings are already available for use, and it won't be long before Windows has a similar offering. Therefore, opt for a CPU with 64-bit support. Also, note that the Athlon64 processors are the fastest at handling 32-bit programs.
- ✓ The CPU speed should depend on the kind of tasks and application the PC is intended for. Do not waste hard-earned money on the fastest CPU to use it for applications that are never going to need that much power. Low-cost variants are available for such tasks.

These will do just fine.

- ✓ Since processors run at high speeds continuously, cooling becomes an important issue. Get a top class heatsink combined with the best cooling fan.
- ✓ Refurbished processors are still available in the market, especially the older generation CPUs that are still in demand in the market. If you ever set out to buy such processors, make sure it comes in an original box pack and has a proper holographic sticker on it.